

Blind Forecast Commitment Market v0.1

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1. Introduction

We propose a type of cryptographic commitment tournament, designed to force unbiased intellectual commitment to a prediction. The idea is to remove herd behavior from prediction markets, and also make it impossible for automated trading bots to profit off of market trends. We can use cryptographic commitments to create a system that allows independent forecasting without social contamination. This system is designed to prevent aggregation and statistical analysis until after resolution. Participants submit private forecasts which are cryptographically committed using a collision-resistant hash function. These commitments are publicly visible while the underlying forecasts remain hidden until after the event resolves. Following resolution, participants may reveal their forecasts and associated nonces, enabling independent verification that the prediction existed before the outcome was known.

2. Motivation

Traditional prediction markets provide useful information aggregation, but they also introduce social effects:

- Herd behavior
- Information cascades
- Strategic signaling

Participants can observe existing market sentiment before forming their own beliefs. The goal of BFCM is different. Instead of measuring: What does the crowd believe? This protocol is designed to extract what each participant independently believes before the outcome occurred. This creates an environment where forecasts are generated in isolation. This in turn removes the advantage of algorithmic trading as well as the advantage of any sort of execution time, as there is not direct incentive to submit predictions at specific times before the predicted on event has passed.

3. Definitions

Let:

- E = Event identifier
- P = Prediction
- N = Random nonce
- $H(\cdot)$ = Cryptographic hash function
- C = Commitment

For Example:

$$E = 123$$

$$P = 1.47$$

$$N = \text{random 128-bit value}$$

A participant creates a commitment by computing:

$$C = H(E || P || N)$$

The commitment C is published publicly. The prediction P and nonce N remain secret. After submitting a commitment, the participant cannot later change P without producing a different hash. This prevents retrospective modification, such as changing of a prediction as a result of observing market behavior and trends. Without knowledge of N , an observer cannot efficiently determine P . The nonce prevents a brute force attack, where an attacker could observe a hash of a blinded prediction that is being widely submitted, and begin to try different inputs until matching the desired hash, subsequently revealing a prediction prematurely. The addition of the nonce method also ensures that every hash observed by participants and organizers is entirely unique. The nonce addition will expand the search space to:

$$2^{128}$$

possible values. This renders brute force attacks computationally infeasible. After the predicted on event has passed, the resolution phase can occur. The participant reveals:

$$(P, N)$$

$$H(E || P || N) = C$$

If equality holds, the forecast is proven to have existed before the event outcome.